Sadashiv Raj Bharadwaj

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Summary

Senior Software Engineer with 8+ years building agentic AI platforms, distributed data systems, and developer infrastructure at scale. Lead architect of **AIDE**, a multi-agent LLM platform serving 17K users and 30 production chatbots on Kubernetes.

Skills

Languages: Python, SQL, Scala, Go, JavaScript, Bash
AI/ML: LangGraph, LangChain, LangSmith, OpenAI, Azure OpenAI, Gemini, Weaviate, RAG, Text-to-SQL, MCP, A2A
Backend: FastAPI, asyncio, Pydantic, SQLAlchemy 2.0, Celery, Redis, gRPC
Infra & Cloud: Kubernetes, Docker, Spark, Kafka, Airflow | GCP (BigQuery, Dataproc, GCS), AWS (EMR, Lambda, S3)
Data & DevOps: Snowflake, PostgreSQL, Hive, Presto | Jenkins, GitHub Actions, Helm, OpenTelemetry

Experience

Cisco Systems — Bangalore, India
Senior Software Engineer, Tech Lead — *Jul 2020 – Present*

AIDE — Generative-AI Agent Platform

- Architected **AIDE 2.0**, a multi-tenant LLM platform (LangGraph, Kubernetes) scaling to **17K users** and **25K concurrent requests**. Designed the core execution engine, API gateway, and GitOps deployment pipeline, supporting **30 production chatbots**.
- Created the **DeepAgent framework**, a skills-based agent runtime with auto-discovered tools, sub-agent delegation, parallel execution, and structured output. Adopted by all 30 production agents; reduced new-bot onboarding from weeks to hours.
- Designed a **typed SSE streaming protocol** and rolled it out across web, Slack, Webex, and mobile clients. Added token-level streaming through human-in-the-loop approval gates, eliminating perceived dead time.
- Built **MCP and A2A interop servers**, exposing every bot as a callable agent via Anthropic's Model Context Protocol and Google's A2A spec. Enabled 6 cross-team agent compositions replacing custom integration code.
- Reduced ad-hoc analyst SQL workload by ~**40%** by shipping a Text-to-SQL agent over 130+ Snowflake tables with column-level catalog search, fiscal-calendar reasoning, and 12 chart types.
- Defined and executed the **platform roadmap** across API, developer experience, observability, and reliability—shipping streaming v2, CLI tooling, distributed tracing, and zero-downtime deploys.
- Led a **team of 6 engineers** across AIDE and Phoenix, driving architecture decisions, code reviews, and design reviews. Mentored 3 junior engineers to independent feature ownership within their first quarter.

Phoenix — Data Orchestration Platform

- Replaced a **$100K/yr licensed tool** by co-architecting Phoenix, a YAML-driven distributed ETL platform that cut data-product delivery from 7 days to 4 hours. Runs 200+ daily Snowflake and BigQuery loads.
- Reduced AWS spend by **$250K/yr** (60% runtime reduction) by engineering multi-threaded, shared-pool pipeline schedulers to replace per-job EMR clusters.
- Built a **self-serve Pipeline Builder UI** (React + FastAPI) adopted by 50+ non-engineering users for authoring and deploying Snowflake and GCP-VM jobs without infrastructure code.
- Built **DACT**, a multimodal GenAI pipeline (ASR + speaker diarization + LLM summarization) that converts executive conference video into transcripts, summaries, and vector embeddings—cutting turnaround from **4–8 hours to under 10 minutes**.
- Deployed a **Data Science Workbench** (Jupyter + Spark + GPU) for 100+ data scientists, reducing ML deployment cycles by 40%.

Gartner — Gurgaon, India
Data Engineer — *Jan 2020 – Jul 2020*

- Migrated legacy ingestion to Apache Airflow on GCP, saving **$65K/yr**; built 40+ high-frequency integrations delivering near-real-time session data into BigQuery.

Nineleaps Technology (contracted to Uber) — Bangalore, India
Software Development Engineer II — *Sep 2017 – Jan 2020*

- Re-engineered Uber's AdTech audience pipeline in PySpark, cutting processing time by **94%** (6 hrs → 20 min) and enabling same-day campaign launches across markets.
- Built a multi-threaded ingestion engine for Uber City Guides, reducing processing latency by **87%** (4 hrs → 30 min).
- Optimized marketing spend allocation by analyzing petabyte-scale rider/driver datasets in Presto and Hive, delivering a measurable **10% lift in marketing ROI**.

Education

University of Delhi — Delhi, India
Bachelor of Technology, Electronics Engineering — *2013 – 2017*